View results

Respondent

26 Anonymous

319:05
Time to complete

A. General information about the respondant

1. * Country

State of Palestine

2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Ministry of Education and Higher Education

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

https://www.moe.edu.ps/

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The Ministry of Education and Higher Education (MoEHE) in Palestine is responsible for overseeing, regulating, and developing the education system at all levels, from early childhood education to higher education and scientific research. Its mandate includes ensuring access to quality education, promoting academic excellence, and aligning educational policies with national development goals.

Key Roles and Responsibilities:

1. General Education (School Education)
- Developing and implementing national education policies, curricula, and standards for pre-school, primary, and secondary education.
- Supervising and supporting public and private schools, ensuring adherence to educational regulations.
- Training and professional development for teachers and school administrators.
- Ensuring inclusive education, particularly for marginalized communities and students with special needs.
- Promoting digital transformation and integrating technology into the education system.
- Overseeing national and international assessment programs to monitor education quality.

2. Higher Education & Scientific Research
- Regulating universities, colleges, and technical institutes, ensuring academic standards and accreditation.
- Developing policies to enhance research, innovation, and scientific collaboration at national and international levels.
- Encouraging partnerships between higher education institutions and industry to align academic programs with labor market needs.
- Supporting scholarships, student financial aid programs, and international academic exchange initiatives.
- Promoting entrepreneurship, innovation, and applied research in universities.

3. Vocational & Technical Education
- Strengthening technical and vocational education and training (TVET) programs to equip students with practical skills for employment.
- Enhancing partnerships with industries and private sector stakeholders to bridge the skills gap.

4. Educational Infrastructure & Quality Assurance
- Constructing and maintaining school and university infrastructure, ensuring safe and modern learning environments.
- Implementing quality assurance mechanisms across all educational institutions.

5. Coordination with International Organizations
- Collaborating with UNESCO, UNRWA, donor agencies, and academic institutions worldwide to improve education policies and secure funding.
- Advocating for Palestinian education rights, particularly in the context of political challenges and restricted access to education.

The MoEHE plays a crucial role in shaping the future of Palestine by investing in human capital, fostering innovation, and ensuring that education remains a key pillar for sustainable development and nation-building.**

7. * Full name of officially designated contact person(s) who completed this survey

Amin Nawahda

8. * Position of officially designated contact person(s) who completed this survey

DG for Development & Scientific Research

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Prof. Amjad Barham

11. Brief description of the consultation process established for the preparation of the report

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Higher education institutions in Palestine, Ministry of Education and Higher Education, Ministry of Telecommunication and digital economy

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

https://mtde.gov.ps/?culture=en-US; https://www.moe.edu.ps/

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- [x] Public
- [ ] Private
- [ ] Civil Society
- [ ] Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- (x) Yes
- ( ) No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Ministry of Education & Higher Education and Higher education institutions

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

- (x) Yes
- ( ) No

19. 1.2 cont. **If yes,** which language(s)?

Arabic and English

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

- (●) Yes
- ( ) No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

The Launching of Palestine researchGate: https://researchgate.ps/#/ , https://researchgate.ps/#/page/PALNREN: Palestinian National Research & Education Network PALNREN Introduction: Scientific research is considered one of the key factors contributing to the advancement of human societies, especially those striving for sustainable development. It is a true measure of the progress and civilization of nations, as it generates knowledge and technology that improve reality. The reputation of research institutions is closely linked to the value and quality of the research they produce, which helps define their leadership at local, regional, and international levels. National Research and Education Networks (NRENs) focus on providing high-quality network connections and related services to universities and research institutions by connecting them with each other and the internet. These networks aim to provide integrated services to students, researchers, educators, and staff at university campuses, as well as to schools, libraries, hospitals, and other public institutions. They also contribute to providing technical support in areas such as security, storage, collaboration, and trust. A research network in any advanced country is considered a key enabler of scientific research and leadership across various fields. Research and education networks aim to build and develop local research capacities in all fields by establishing a national high-speed communication network that offers advanced information services, stimulating researchers to increase their research activities and enhancing the quality of their output. Additionally, these networks work to unify efforts in distributing educational resources and optimizing financial resources, while expanding to open doors to similar global networks, enabling local researchers to connect with international peers and participate in research resources and services offered by other universities locally and internationally, such as high-performance computers, cloud services, joint programs, and quantitative research results. Some of the services provided by research and education networks include video conferencing, lecture and conference streaming over the network, facilitating direct communication between researchers, and easy data transfer for their projects without the need for travel or delays caused by traditional data transfer methods. They also offer educational roaming (EDU Roam), allowing researchers to move between universities within the same country or across institutions sharing the same service. Furthermore, some research and education networks offer a unified subscription to global digital libraries and central library systems. They also provide services for student information systems (SIS), learning management systems (LMS), enterprise resource planning (ERP), and other applications for managing university campuses and educational processes in the country. These networks aim to train national experts in network management, create virtual scientific communities, and enrich the community with the latest knowledge releases, offering local researchers the opportunity to directly communicate with global experts in various fields. Vision of the Network: The Palestinian Research Network aims to empower innovation, scientific research, and education in Palestine through local and international strategic collaboration between the research and educational communities. The network seeks to establish a strategic partnership with the Palestinian Telecommunications Company (Paltel) and other telecom service providers to build, operate, and maintain this network. It also strives to support its members in utilizing and producing knowledge, increasing research capabilities, and innovating new ideas, patents, and products, thus advancing scientific research activities in Palestine and enhancing Palestine's leadership role in the region. Mission of the Network: To enable a sustaina

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

- [x] Yes
- [ ] No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Yes, specific actions have been undertaken to incorporate the values and principles of open science in publicly funded research in Palestine. One key initiative is the ResearchGate.ps portal, launched by the Ministry of Education and Higher Education. This platform enables researchers in Palestine to access university repositories, join scientific groups, and share published papers and research projects. The portal supports open scientific knowledge by providing open access to research publications, data, and scientific collaborations. It also contributes to open science infrastructure by facilitating knowledge-sharing among researchers, students, and institutions. Furthermore, ResearchGate.ps encourages open engagement of societal actors by allowing broader participation in research discussions and fostering transparency in scientific communication. Future efforts will continue to enhance open science practices, ensuring alignment with UNESCO's 2021 Recommendation on Open Science.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

- (●) Yes
- ( ) No

25. 2.1 cont. **If yes**

· please share a machine-readable PDF file of the policy using this link: **https://tinyurl.com/27p5zct8** designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- [ ] open access to scientific knowledge
- [ ] open science infrastructure
- [x] open engagement of societal actors
- [x] open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ○ Yes
- ○ Partly
- ◉ No

28. 2.2 cont. **If no,** please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

Currently, there is no national policy or set of policies specifically addressing open science or its key elements in line with the 2021 Recommendation on Open Science. The main obstacles include a lack of awareness and understanding of open science principles among policymakers, insufficient funding for open science initiatives, and the absence of a legal framework to support open data sharing, open access publishing, and collaborative research. Additionally, there is limited infrastructure for implementing open science practices, and existing research policies do not explicitly address the need for open science. There are discussions within certain academic and scientific circles regarding the development of such policies, but concrete plans and actions are still in early stages

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ( ) Yes
- ( ) Under development
- (•) No

30. 2.3 cont. **If no,** please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

At present, there are no specific policy instruments in place to promote open science in the country. While there may be individual initiatives within academic institutions or research organizations, these are not part of a national or coordinated effort. The absence of dedicated programmes, methods, or operational mechanisms to address open science challenges means there is a lack of structured support for advancing open data, open access, and collaborative research practices. This gap is compounded by limited resources allocated to such initiatives, and the lack of clear indicators or goals for evaluating their effectiveness.

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? *(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)*

- (•) Yes
- ( ) Under development
- ( ) No

32. 2.4 cont. **If yes or under development,** please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

A national implementation plan for open science is currently under development in Palestine. This is being coordinated by the Palestinian National Commission for Education, Culture, and Science, which recently held a specialized workshop on "Open Science Practices in Palestine." The workshop aimed to discuss the UNESCO Recommendation on Open Science and its related principles. The workshop involved various key stakeholders, including: Dr. Duwas Duwas, Secretary-General of the National Commission and President of the Executive Council of ISESCO Rania Jaber, Director-General of the Technology Innovation and Creativity Center at the Ministry of Communications and Information Technology Mahmoud Hawamda, Director of the Digital Education Center at Al-Quds Open University Dr. Amin Nawahda, Deputy Director-General of Development and Scientific Research at the Ministry of Higher Education and Scientific Research Dr. Majdi Ouda, holder of the UNESCO Chair in Data Science at the Arab American University The discussions emphasized the importance of Palestine's participation in global scientific networks to encourage Palestinian researchers' access to open resources. The National Commission is working towards formulating a national framework for open science, coordinating with relevant institutions to define national needs and capacities. A key recommendation from the workshop was the formation of a local open science working group and the establishment of mechanisms for international collaboration. This group will draft a report on Palestine's progress in open educational resources, which will be submitted to international organizations by the end of the month.

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? *(ref.: (ii) 17.d, 17.e, 17.g)*

- ○ Yes
- ○ Under development
- ● No

34. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? *(ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)*

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ○ Yes
- ○ Under development
- ◉ No

35. 2.6 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

Currently, there are no specific funding mechanisms at the national level dedicated to open science in the country. The primary funding sources for research are generally directed toward specific scientific projects or institutions, without particular focus on open science initiatives such as open data, open access publishing, or infrastructure for open science. The absence of dedicated funding mechanisms for open science is partly due to a lack of awareness about the importance of such initiatives and insufficient alignment of national research funding priorities with open science principles. Additionally, there is a limited budget for innovation, digital infrastructure, and science communication, which hinders the development of sustainable open science practices. However, there are ongoing discussions to explore the potential for creating dedicated funding mechanisms for open science, and efforts are being made to align national funding strategies with global open science trends.

36. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

- (selected) Yes
- No

37. 2.7 cont. **If yes**, please provide details and links as relevant.

(https://researchgate.ps/#/page/repositories)Research Repositories in Palestinian Universities Research repositories in Palestinian universities represent one of the fundamental pillars for promoting scientific research and disseminating knowledge. These repositories provide access to scientific studies, theses, articles, and other academic materials produced by universities. They also contribute to enhancing academic collaboration among researchers locally and internationally. Below is a list of research repositories in Palestinian universities along with their access links: Islamic University – Gaza IUGSpace An-Najah National University Scholar Najah Al-Azhar University – Mishkat Birzeit University – FADA Palestine Polytechnic University Scholar PPU Palestine Technical University – Kadoorie Scholar PTUK Al-Quds Open University – Osoul Al-Aqsa University Scholar Al-Aqsa University of Palestine DSPACE UP Al-Istiqlal University – Tamayouz Al-Quds University DSPACE Al-Quds Arab American University – Wasm Hebron University DSPACE Hebron Bethlehem University DSPACE Bethlehem University College of Applied Sciences DSPACE UCAS Importance of Research Repositories Promoting Open Access: Enhancing open dissemination of scientific knowledge. Improving Accessibility: Facilitating access to academic and research materials. Supporting Sustainability: Advancing academic efforts to achieve sustainable development.

38. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

- ○ Yes
- ◉ No

39. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Currently, there have been no significant efforts or plans to foster equitable public-private partnerships specifically focused on open science in the country. The development of such partnerships faces several challenges, including limited awareness of the importance of open science in both public and private sectors, a lack of clear incentives for private companies to engage in open science practices, and the absence of formal frameworks to encourage collaboration in this area. Additionally, the public sector has limited capacity and resources to engage with private companies on open science initiatives, and private sector players may not yet see the direct benefits of contributing to open access, open data, and collaborative research practices. However, there is potential for future development, and some discussions have been initiated on how public-private collaborations could help bridge the gaps in open science, especially if funding and policy support are improved. These discussions may lead to concrete actions by the end of 2025.

40. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ○ Yes
- ○ Under development
- ◉ No

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

41. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

NA

42. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

As of 2023, approximately 86.64% of the Palestinian population has access to the internet, reflecting a slight decline from 88.65% in 2022. THEGLOBALECONOMY.COM In 2022, about 92% of Palestinian households reported having internet access at home, with 93% in the West Bank and 92% in the Gaza Strip. ENGLISH.WAFA.PS Additionally, 89% of individuals aged 10 and above used the internet from any location in 2022, with 92% in the West Bank and 83% in the Gaza Strip. ENGLISH.WAFA.PS These figures indicate a high level of internet penetration in Palestine, with slight variations between regions.

43. 3.3 Are there national research and education networks (NRENs)[i] active in the country? *(ref.: (iii) 18.c)*

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

- (●) Yes
- ( ) No

44. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

https://researchgate.ps/#/page/PALNREN

45. 3.4 Are there national or institutional open science infrastructures[i] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- (selected) Yes
- Under development
- No

46. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- [ ] federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- [ ] community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- [x] platforms for exchanges and co-creation of knowledge between scientists and society
- [ ] community-based monitoring and information systems to complement national, regional and global data and information systems

47. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

https://researchgate.ps/#/

48. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

- ( ) Yes
- (•) No

49. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

- [x] Yes
- [ ] No
- [ ] Other

50. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

- ○ Yes
- ◉ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

51. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

- ○ Yes
- ◉ No

52. 4.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Despite the growing recognition of the importance of open science, systematic capacity-building initiatives in higher education and research-performing institutions in Palestine have not yet been fully established. The main challenges include: Limited Institutional Frameworks – There is no comprehensive national policy or strategy that mandates open science practices, leading to inconsistent efforts across institutions. Funding Constraints – The lack of dedicated financial support for open science training and infrastructure hinders the ability of universities and research centers to develop systematic capacity-building programs. Technical and Infrastructure Barriers – Limited access to advanced digital repositories, data-sharing platforms, and high-speed internet in some areas makes the implementation of open science practices challenging. Awareness and Cultural Shifts – Many researchers and institutions still follow traditional publishing and data-sharing models, requiring efforts to shift mindsets and promote the benefits of open science. However, there are discussions and ongoing efforts to integrate open science training into research and academic programs. Future plans include collaborative initiatives with international organizations, potential funding proposals, and the development of national guidelines to facilitate capacity-building in this area.

53. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

- ○ Yes
- ◉ No

54. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In Palestine, capacity-building programs specifically designed for policymakers on integrating the core values and principles of open science into Science, Technology, and Innovation (STI) policies and strategies have not yet been systematically implemented. Challenges and Obstacles: Absence of a National Open Science Strategy – There is no formalized national framework that mandates training for policymakers on open science, leading to a lack of structured capacity-building efforts. Limited Financial and Technical Resources – Funding constraints and the absence of dedicated resources hinder the development and execution of specialized training programs. Competing Priorities – Policymakers are often engaged with pressing national issues, and open science capacity-building has not yet been prioritized at the policy level. Lack of Awareness and Expertise – Many policymakers may not be fully aware of the benefits of open science or how to integrate it into STI policies effectively. Future Plans: There is growing interest in aligning Palestine’s STI policies with international open science recommendations. Efforts are being made to engage with international organizations such as UNESCO and ALECSO to explore potential training programs, workshops, and policy dialogues to equip policymakers with the necessary knowledge and skills to integrate open science principles into national STI strategies.

55. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

- (selected) Yes
- No

56. 4.3 cont. **If yes**, please provide more information and links as relevant.

https://researchgate.ps/#/page/repositories

57. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? *(ref.: (iv) 19.d)*

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ○ Yes
- ○ Partly
- ◉ No

58. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ◉ Yes
- ○ No

59. 4.5 cont. **If yes**, please provide more information and links as relevant.

The Palestinian Research Gate aims to strengthen the research infrastructure and enhance academic collaboration within Palestine and beyond. The Gate provides an interactive digital environment that supports researchers in managing and publishing their research, facilitating access to research resources and advanced technologies. It also aims to promote transparency and access to scientific data, enhancing researchers' capabilities through training courses and capacity building. The Gate is committed to supporting national research priorities and fostering integration between academic, research, and industrial institutions to achieve sustainable development and innovation in various fields that serve the scientific and economic community in Palestine. Additionally, the Palestinian Research Gate seeks to foster a culture of research and innovation in the Palestinian community, strengthening ties between researchers and local and international institutions. The Gate aims to create an encouraging environment for learning and innovation, motivating young people to pursue successful academic and research paths, contributing to the development of human resources and enhancing the knowledge economy in Palestine.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

'

60. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: (v) 20.b, 20c, 20i)*

- ○ Yes
- ◉ No

61. 5.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In Palestine, there have not been systematic national or institutional initiatives to review existing research assessment and evaluation processes specifically to align them with open science principles and values. Challenges and Obstacles: Traditional Research Assessment Models – Current evaluation processes prioritize conventional metrics such as journal impact factors and citation counts, with limited emphasis on open science practices like open access publishing and data sharing. Lack of Institutional Policies – Universities and research institutions have yet to adopt formal policies integrating open science into research assessment frameworks. Resource and Capacity Constraints – Limited funding and expertise make it challenging to implement reforms in research evaluation that support open science. Awareness and Cultural Barriers – Many researchers and institutions are accustomed to traditional assessment criteria, and there is resistance to shifting towards open science-aligned evaluation systems.

62. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

- ○ Yes
- ◉ No

63. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

64. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

- ○ Yes
- ◉ No

65. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

66. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

- [ ] Yes
- [x] No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

67. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

- ○ YES, at the national level
- ○ YES, at the institutional level
- ○ YES, at both the national and institutional levels
- ◉ No

68. 6.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

69. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Palestine is actively promoting and facilitating international scientific collaborations on open science in alignment with the 2021 Recommendation on Open Science through several key initiatives: 1. National Open Science Community A national community for open science exists in Palestine, which regularly engages with UNESCO and other international organizations to share information, best practices, and updates on open science developments. This platform helps connect Palestinian researchers with global networks, fostering knowledge exchange and collaboration. 2. Membership in ASREN Palestine is a member of the Arab States Research and Education Network (ASREN), which enhances its integration into regional and global research and education networks. Through ASREN, Palestinian researchers benefit from improved digital infrastructure, knowledge-sharing, and collaborative opportunities with international partners. 3. Participation in International Programs Palestinian universities and research institutions actively participate in global open science initiatives, such as UNESCO-led programs and regional collaborations. These engagements help integrate Palestinian research into the global scientific community, encouraging joint research projects, data-sharing initiatives, and cross-border academic exchange. 4. Open Science Capacity Building Workshops, training sessions, and knowledge-sharing events are regularly organized to raise awareness among researchers, policymakers, and academic institutions about open access publishing, open data sharing, and collaborative research methodologies. This ensures that Palestinian scientists are equipped to engage in international open science collaborations effectively. 5. Strengthening Institutional Policies Palestinian academic and research institutions are increasingly aligning their policies with open science principles, facilitating joint research with international partners and ensuring compliance with global standards for open-access publications and data-sharing protocols. 6. Digital Infrastructure Development Efforts are being made to enhance digital repositories and open-access platforms that support international research collaborations. These platforms provide Palestinian researchers with opportunities to contribute to and benefit from global knowledge networks.

70. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

- ○ Yes
- ◉ No

71. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

72. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

- (selected) Yes
- No

73. 7.3 cont. **If yes**, please provide links as relevant and more information.

https://connect.geant.org/2024/09/09/eumedplus-strengthening-digital-connectivity-and-innovation-in-the-eus-southern-neighAlgeria, Egypt, Jordan, Lebanon, Libya, Morocco, Palestine, Syria and Tunisia as a group are part of what is known as the European Union's Southern Neighbourhood countries. With their predominantly young population experiencing significant unemployment rates, there is an urgent need for investment in human capital to foster economic growth, create societal opportunities, and mitigate brain drain. Enter: the research and education (R&E) sector. R&E networks are crucial in this effort, particularly through digital transformation, which enables access to high-speed internet, data infrastructures, and e-services that are essential for collaboration, innovation, and evidence-based decision-making For this reason and building on the successes of previous EU-funded projects – such as AfricaConnect and EUMEDCONNECT, the EU co-funded EUMEDplus project was launched on 1 September 2024. Aligned with the Agenda for the EU's partnership for the Southern Neighbourhood, EUMEDplus aims at enhancing the capabilities of NRENs in North Africa and their Regional Network, the Arab States Research and Education Network (ASREN), and increasing access to modern digital infrastructure and technologies. With a four-year timeline and a budget of 13,334,000 euros – of which 12,000,000 euros are contributed by the European Union – the project is coordinated by GÉANT with key partners including ASREN, the American University of Beirut, the Centre National pour la Recherche Scientifique et Technique in Morocco, the Cyprus Research and Academic Network, NORDUnet, and SESAME in Jordan. The partners will work collaboratively to ensure that the project's objectives are met and that its impact is felt across the Southern Neighbourhood. The project is designed to address key challenges in digital connectivity in the EU's Southern Neighbourhood as a catalyst for human capacity development. EUMEDplus aims to achieve this ambition by: Supporting Research and Education: in particular, by strengthening the role of ASREN and North African NRENs in driving digital transformation within the R&E sector. The project aims to enable them to participate more fully in global scientific collaboration and innovation by enhancing their maturity and sustainability development as well as operational efficiency, both at regional and national level. It will also prioritise gender equality-oriented initiatives through dedicated conference tracks, mentorship programmes and training courses. Elevating Digital Infrastructure: in order to significantly improve access to modern digital infrastructures and technologies, the project will aid North African NRENs in developing and implementing national strategies to fully utilise the pan-Mediterranean Medusa submarine-cable project, EU's first digital global Gateway undertaking, and also investigate the feasibility of expanding the Medusa system, now between North Africa and Europe, towards the Levant, whist ensuring ongoing international R&E connectivity for the Levant. Expanding the service offering: the project will empower NREN to tackle the challenge of making data infrastructure scalable and value-added services and tools that are accessible to end users. The focus will be on establishing the underlying federation systems for secure service access, fostering cross-border cooperation in HPC utilisation and promoting the adoption of open science principles. EUMEDplus is set to play a transformative role in the Southern Neighbourhood by enhancing digital infrastructure, supporting educational and research institutions, and fostering sustainable, inclusive growth. Through its focus on long-term sustainability, gender equality and high-speed connectivity, the project promises to bridge the digital divide and stimulate innovation, ensuring that the Southern Neighbourhood is well-positioned to thrive in the digital age.

Part 8. GENERAL CONSIDERATIONS

74. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

- (●) Yes
- ( ) No

75. 8.1 cont. **If yes**, please provide more information.

It should be recognized that Palestine is still under the Israeli occupation and will suffer for long due to the war in Gaza

76. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org